GenFed-IDS: A Lightweight Federated Generative AI Framework for UAV Anomaly Detection in Rescue Operations

Hafiz Muhammad Attaullah, Muhammad Ehsan, Shakila Basheer, Ala Saleh Alluhaidan

***Abstract*—Unmanned Aerial Vehicles are increasingly deployed in consumer applications such as logistics, disaster recovery, and surveillance, yet their wireless communication links remain highly vulnerable to cyber-attacks. Traditional intrusion detection systems (IDS) often struggle in UAV environments due to resource constraints, dynamic network conditions, and scarcity of labeled datasets. To address these challenges, we propose a Generative AI-enabled lightweight IDS framework tailored for UAV communication networks. The framework integrates hybrid Convolutional Neural Network–Gated Recurrent Unit (CNN-GRU) autoencoders with generative augmentation and knowledge distillation, achieving high accuracy while maintaining computational efficiency. Explainability is incorporated via SHapley Additive exPlanations (SHAP) analysis to ensure trustworthy and interpretable decision-making. Experimental evaluations on a multimodal UAV dataset demonstrate state-of-the-art performance with 99.49% accuracy, 99.48% recall, and AUROC of 0.9999, alongside a student model that reduces model size, inference latency, and memory footprint by nearly 50%. Comparative results confirm that the proposed framework outperforms recent IDS baselines in both detection capability and lightweight deployment, offering a practical and scalable solution for next-generation UAV communication networks.**

***Index Terms*—IDS, UAV, Generative AI, Federated Learning, Variational Autoencoder, Cyber-Physical Security.**

I. Introduction

Unmanned Aerial Vehicles (UAVs) are rapidly transforming modern communication and consumer electronics ecosystems. From smart city surveillance and disaster recovery to logistics and agriculture, UAVs are increasingly deployed in critical and consumer-centric applications [1]. These UAV systems rely heavily on wireless communication links for command-and-control, navigation, and cooperative tasks.

As illustrated in Fig. 1, a UAV-assisted rescue system typically consists of a UAV platform equipped with high-resolution cameras, first-aid payloads, and communication modules. The UAV communicates securely with the Ground Control Station (GCS) via a dedicated wireless link, ensuring real-time command-and-control as well as data transmission. Simultaneously, the UAV interacts with rescue payloads and on-site devices such as first-aid kits and sensors through direct communication links. The GCS, in turn, coordinates with the central rescue command center, which oversees mission planning and operational decisions. This architecture enables rapid situational awareness, efficient coordination, and timely delivery of emergency services in disaster-stricken or hard-to-reach environments.

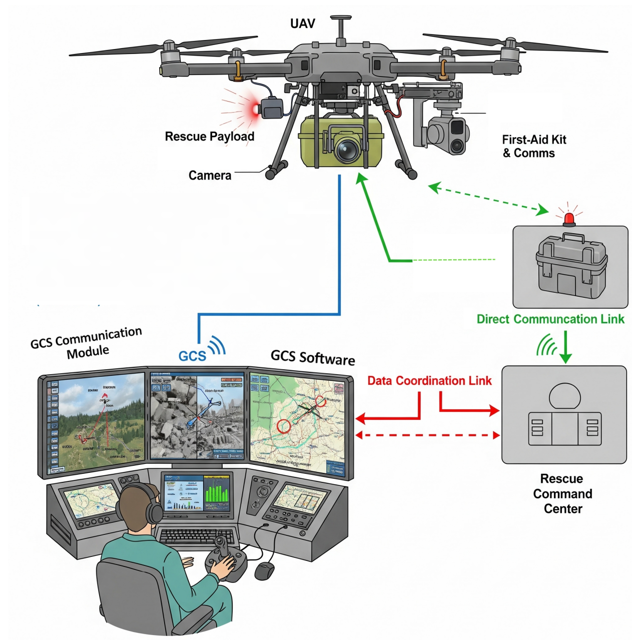

Fig. 1: Architecture of UAV-assisted rescue operations showing communication links between UAV, Ground Control Station, rescue payload, and command center.

However, their open wireless medium and distributed architecture make them highly vulnerable to cyber-attacks [2]. UAV systems are vulnerable to a broad spectrum of adversarial threats to both cyber and physical layers. On the cyber level, the adversaries can use communication channels to interfere with the telemetry and mission data by jamming, hacking, or spoofing. Physically, UAVs may be tampered with or their payload altered. Moreover, data security and operational security are also important because mission-critical information should be confidential, authentic, and tamper-resistant throughout rescue operations. The physical attacks to cyber and data-oriented threats hierarchical nature of UAV threat models requires effective security solutions that can be adapted

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The authors extend their appreciation to the Deanship of Scientific Research and Libraries in Princess Nourah bint Abdulrahman University for funding this research work through the Program for Supporting Publication in Top-Impact Journals, Grant No. (SPTIF-2025-32).

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to changing environments and resource-limited UAV platforms [3].

Secure and trustworthy UAV communication is thus essential in ensuring safe integration of UAVs in consumer and industrial ecosystems. One of the effective defenses against such attacks has been the Intrusion Detection Systems (IDS) that monitors the communication traffic and detects anomalies in it [4]. In this regards, the main protection of this study is the communication and network level UAV channels against jamming, spoofing, and packet-level manipulation, and the reduction of the effects of compromised FL clients.

However, traditional IDS systems have major limitations in UAV networks because of resource limitations, high mobility, dynamic topology, and lack of attack datasets. Conventional ML- and DL-based IDS methods are based on large labeled datasets, fixed signatures, or complicated deep models, which are not applicable to UAV networks. UAV-specific data is limited and unbalanced, UAVs are in a highly dynamic wireless environment, and onboard devices have severe energy and computation constraints. In addition, most IDS systems are black boxes, which are not interpretable, which is a requirement of mission- and safety-critical UAV operations.

Generative AI (GenAI) solves these issues with the help of Variational Autoencoder (VAE), generative adversarial networks (GANs), diffusion models, and lightweight Transformers. It is capable of generating realistic attack traffic to alleviate the problem of data scarcity, train compact latent features to effectively detect anomalies, and adapt to unknown attacks through generative-discriminative training. GenAI also enables explainable IDS, revealing abnormal patterns, which makes systems reliable and deployable in UAV settings.

This paper presents a Generative AI-based Intrusion Detection Framework that is specific to UAV communication networks. An edge-deployed lightweight anomaly detection model is trained using quantization and pruning to be energy-efficient.

The major contributions of this paper are:

- We propose a **GenAI-enabled IDS** for UAV networks, integrating lightweight VAE-based representation learning with GAN-driven data augmentation.
- We develop a **resource-efficient detection pipeline** using pruning, quantization, and knowledge distillation for real-time UAV deployment.
- We enhance **trustworthiness via explainability**, employing SHapley Additive exPlanations (SHAP) and attention-based insights for interpretable decisions.
- We conduct **comprehensive evaluations**, showing higher accuracy, lower latency, and reduced energy use than existing ML/DL baselines.

The remainder of the paper is organized as follows: Section II reviews related work; Section III details the proposed methodology; Section IV presents the experimental setup; Section V discusses results; and Section VI concludes the paper.

II. Related Work

There are three broad classes of IDS, namely: *signature-based*, *anomaly-based*, and *hybrid*. Signature-based IDS are based on pre-defined patterns or attack signatures to detect malicious activities and are therefore effective in known threats but not in zero-day or new attacks. Anomaly-based IDS observe system behavior or communication patterns and indicate anomalies in behavior as possible intrusions. These systems can identify new and emerging attacks and evolving attacks, but they are usually characterized by high false alarm rates because of dynamic environments. Hybrid IDS are a combination of the two methods that use signature databases and anomaly detection modules to enhance accuracy and resilience, but these systems are generally more resource-intensive.

Recent research has investigated the design of IDS in the IoT, UAVs, and Industry 5.0 application with growing interest in explainability, data privacy, and generative models. Indicatively, the article in [5] introduced a federated multi-party computation (FMPC)-based collaborative CNN model to perform privacy-aware pneumonia diagnosis, which showed high accuracy and maintained data confidentiality. Though the method is mostly aimed at healthcare IoT, the role of federated and privacy-preserving learning is emphasized in the context of UAV communication systems. On the same note, [6] created a explainable and robust IDS by integrating BiLSTM and BiGRU models with SHAP-based feature attribution, which is highly robust to poisoning attacks. Although this research offers a basis to reliable IDS, its analysis is confined to conventional IoT data without UAV-specific modification.

Within the context of consumer IoT security, a new framework was proposed in the article by [7] that combines off-policy Proximal Policy Optimization (PPO) reinforcement learning with GAN-based augmentation of DDoS detection in healthcare IoT systems. This method showed better flexibility in changing conditions, yet its healthcare-specific dataset shows that it is necessary to adapt to UAV conditions. In [8] a strong generative collaborative IDS was suggested, which integrates federated learning (FL) with differential privacy and generative models, demonstrating encouraging outcomes on heterogeneous edge IoT data. Nevertheless, the UAV telemetry and multi-modal communication data were not tackled completely. Equally, a study by [9] explored a clustering-based anomaly detection system that is improved with multi-critic GANs to semi-supervised IDS to facilitate the successful detection of rare attacks in semi-labeled datasets. This method is particularly applicable to UAV traffic in which labeled datasets are scarce, but UAV-specific validation has not been studied.

Another emerging line of work focuses on adversarial robustness and explainability. In [10], the authors presented NonsaliencyCrossover, an explainable AI-driven adversarial example generation strategy for IDS, where SHAP-guided perturbations were used to evaluate and strengthen model robustness. While promising, the study remains constrained to conventional IoT datasets, highlighting the opportunity to extend adversarial robustness research to UAV-based IDS.

A trustworthy intrusion detection system leveraging explainable AI was built for in-vehicle networks in [11], where standard deep learning models displayed modest accuracy with better interpretability and robustness. Subsequently, a GAN-

TABLE I: Summary of Key IDS Approaches Relevant for Generative Federated UAV and IoT Security

Ref.	Model / Approach	Dataset Used	Key Focus	Limitations
[16]	FedAvg, DNN	X-IIoTID, edge IIoT-set, WUSTL-IoT-2021	Federated IDS with high accuracy, privacy, and scalability	FL vulnerable to poisoning attacks, limited explainability for UAV/IoT settings
[17]	FedGen+, Generative Fed Distillation, GANs	Edge/IoT datasets	Handles non-IID, robust generative distillation	Requires strong aggregation, limited UAV evaluation, mostly IoT focus
[18]	FedGAN, WGAN-GP	IVN/CAN bus datasets	FL GAN for privacy-preserving attack sample generation	Focus on IVN, limited UAV or multi-modal IoT applicability
[19]	FFCNN collaborative DL IDS	UAVIDS	Real-time collaborative UAV-IDS, zero-day attack detection	Missing generative augmentation, lacks explainability components
[8]	DP-GAN, FL	Edge-IIoTset, CIC, TON-IoT	FL + GAN with differential privacy for collaborative IDS	Less explored in resource-constrained UAV-edge environments
[20]	Human-in-the-Loop + GAN	UAV network traffic	Combines HITL and GAN to improve IDS	HITL scalability issues, needs constant human feedback
[21]	GAN-based adversarial training	UAVIDS, IoFT attack traces	Robust adversarial UAV-IDS via GAN augmentations	Lacks formal FL privacy validation, attack scope narrow
[22]	Generative Few-Shot Learning (CDDPM, CNNBiGRU)	CICIDS2017, CICD-DoS2019	Few-shot generative rare-class detection	Not UAV-specific, no integrated federation or GenAI pipeline
[23]	CNN-LSTM hybrid, IoMT edge IDS	CSE-CIC-IDS2018	Real-time and edge-friendly blended DL IDS	Lacks GenAI and federated learning, focused on IoMT only
[24]	Conceptual Survey (GenAI, Fed, Semantic Comms)	UAVNet, UAVIDS, IoT UAV datasets	Reviews emerging GenAI, FL, and challenges in UAV/IoT IDS	High-level overview, limited experimental UAV results

based IDS targeting automobile CAN networks was proposed in [12], obtaining increased detection rates by adversarial sample augmentation and deep feature extraction. Cloud-based cybersecurity solutions harnessing CNN architectures were given in [13], concentrating on remote detection in automotive contexts with balanced computational costs and detection effectiveness. Collaborative blockchain-enabled IDS frameworks boosting IoT and cloud network security while preserving privacy were introduced in [14], merging anomaly detection with distributed ledger technology. Lastly, a hybrid semi-supervised GRU model was applied for anomaly detection in vehicular networks in [15], demonstrating effective feature representation and elevated accuracy in challenging traffic settings.

A broader survey of related IDS approaches is summarized in Table I. Several frameworks have integrated federated learning with GANs and differential privacy to enhance scalability, data protection, and robustness across distributed IoT environments [8], [16]–[18]. Collaborative UAV-specific IDS models, such as FFCNN-based architectures, have been proposed to support real-time detection and zero-day attack identification [19], though explainability and generative augmentation are still lacking. Other studies have introduced Human-in-the-Loop IDS enhanced by GAN-generated synthetic samples [20], and adversarial training strategies to improve UAV IDS robustness [21]. Furthermore, few-shot generative learning approaches [22] and CNN-LSTM hybrid IDS frameworks [23] have advanced anomaly detection in IoT networks but remain resource-intensive for UAV deployment.

Finally, conceptual surveys [24] highlight the promise of generative AI and federated learning in UAV and IoT security but do not offer much experimental validation.

Overall, while recent works demonstrate significant progress in IDS design for IoT and UAV networks, challenges remain in addressing data scarcity, dynamic UAV communication environments, adversarial robustness, and explainability. This motivates our proposed Generative AI-enabled IDS framework that specifically targets UAV communication networks with a lightweight, energy-efficient, and interpretable design.

III. Proposed Methodology

The proposed Generative AI-enabled IDS framework for UAV communication networks is illustrated in Fig. 2. The system is composed of four stages: (i) data preprocessing, feature engineering, and cross-validation setup; (ii) teacher–student learning using a hybrid CNN–GRU autoencoder distilled into a lightweight model; (iii) evaluation through federated learning, explainable AI, and ablation studies; and (iv) export and deployment for UAV applications. This design ensures resource efficiency while maintaining robustness against evolving threats.

First we design a lightweight Variational Autoencoder as the core of our Generative Encoder Module (GEM). Let $\hat{x}t \in \mathbb{R}^d$ denote the normalized feature vector corresponding to a UAV traffic window. The encoder network $f\phi : \mathbb{R}^d \to \mathbb{R}^{2k}$ projects $\hat{x}t$ into a mean vector $\mu\phi(\hat{x}t) \in \mathbb{R}^k$ and a log-variance vector $\log \sigma^2\phi(\hat{x}_t) \in \mathbb{R}^k$ which parameterize the approximate posterior distribution as in Eq. (1) and 2:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}, \tag{1}$$

$$q_\phi(z_t|\hat{x}t) = \mathcal{N}\Big(z_t;, \mu\phi(\hat{x}t),, \mathrm{diag}(\sigma^2\phi(\hat{x}_t))\Big). \tag{2}$$

Latent samples are obtained via the reparameterization trick as Eq. (3):

$$z_t = \mu_\phi(\hat{x}t) + \sigma\phi(\hat{x}_t) \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I_k), \tag{3}$$

which enables gradient backpropagation through stochastic sampling. The decoder $g_\theta : \mathbb{R}^k \to \mathbb{R}^d$ reconstructs the input by generating a probability distribution over $\hat{x}_t$, as shown in (4):

$$p_\theta(\hat{x}_t|z_t) = \mathcal{N}\big(\mu_\theta(z_t), \sigma_x^2 I_d\big). \tag{4}$$

The VAE is optimized by minimizing the negative Evidence Lower Bound (ELBO), given in (5):

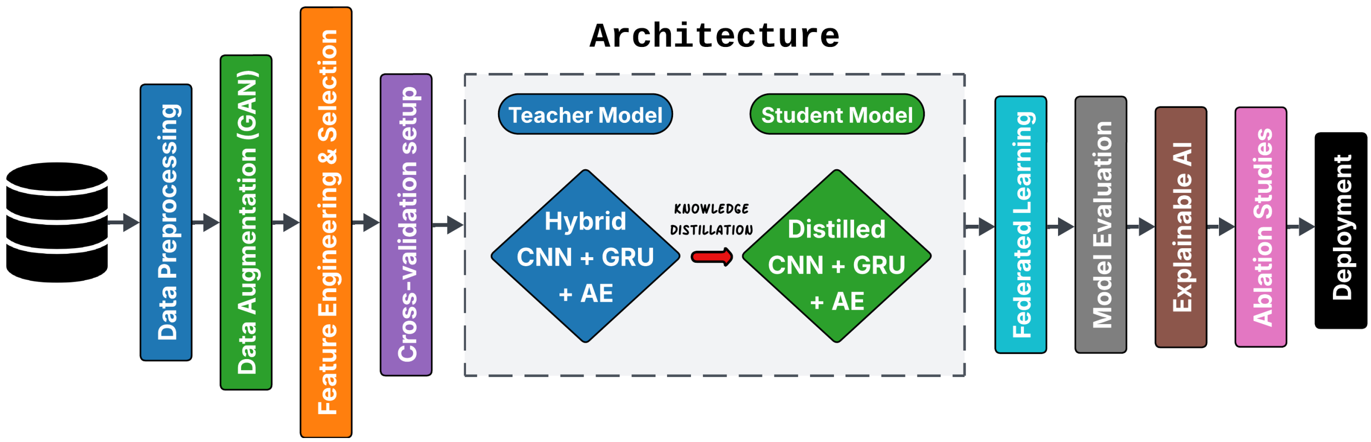

Fig. 2: Comprehensive methodology pipeline of the proposed framework, highlighting the steps.

$$\mathcal{L}_{VAE} = \mathbb{E}_{q_\phi(z|\hat{x})}\Big[\|\hat{x}_t - \mu_\theta(z_t)\|_2^2\Big] + \beta\, D_{KL}\big(q_\phi(z|\hat{x}_t)\,\|\,p(z)\big), \tag{5}$$

where $p(z) = \mathcal{N}(0, I)$ is the isotropic Gaussian prior and D_{KL} represents the Kullback–Leibler divergence. The explicit form of the KL divergence is expressed in (6) and compression in (8):

$$D_{KL}\big(q_\phi(z|\hat{x}_t)\,\|\,p(z)\big) = \frac{1}{2}\sum_{j=1}^{k}\left(1 + \log\sigma_j^2 - \mu_j^2 - \sigma_j^2\right). \tag{6}$$

$$\begin{aligned}\mathcal{L}_{GAN} = {} & \mathbb{E}_{x\sim p_{\text{data}}}[\log D(x)] \\ & + \mathbb{E}_{z\sim p_z(z),\, y\sim p(y)}\big[\log(1 - D(G(z \mid y)))\big].\end{aligned} \tag{7}$$

$$\theta_{\text{compressed}} = \mathcal{Q}_{\text{INT8}}\big(\mathcal{P}_\kappa(\theta_{\text{teacher}})\big)\,, \tag{8}$$

Unlike conventional VAEs, our GEM integrates a supervised contrastive loss that improves latent space discriminability for IDS tasks. Specifically, given a batch of embeddings $\{z_1, \ldots, z_B\}$ with labels $\{y_1, \ldots, y_B\}$, we enforce that embeddings of the same class are pulled closer, while embeddings of different classes are pushed apart, as defined in (9):

$$\mathcal{L}_{con} = \sum_{i=1}^{B}\frac{-1}{|\mathcal{P}(i)|}\sum_{p\in\mathcal{P}(i)}\log\frac{\exp(\text{sim}(z_i, z_p)/\tau)}{\sum_{a=1}^{B}\exp(\text{sim}(z_i, z_a)/\tau)}, \tag{9}$$

where $\mathcal{P}(i)$ is the set of positive indices sharing the same label as i, $\text{sim}(z_i, z_j) = \frac{z_i^\top z_j}{\|z_i\|\|z_j\|}$ is cosine similarity, and τ is a temperature parameter controlling separation sharpness.

The complete GEM objective is thus defined in (10):

$$\mathcal{L}_{GEM} = \mathcal{L}_{VAE} + \lambda\mathcal{L}_{con}, \tag{10}$$

with λ balancing reconstruction fidelity against discriminative structure. This ensures that the learned latent codes is compact for deployment on UAV and sufficiently structured to enable downstream anomaly detection.

$$f(z_t) = \text{sign}\left(\sum_{i=1}^{N}\alpha_i \exp\big(-\gamma\|z_t - z_i\|_2^2\big) - \rho\right), \tag{11}$$

$$\begin{aligned} f(x) &= \varphi_0 + \sum_{j=1}^{d}\varphi_j\, x_j, \\ \varphi_j &= \sum_{S\subseteq\{1,\ldots,d\}\setminus\{j\}}\frac{|S|!\,(d - |S| - 1)!}{d!}\big(f_{S\cup\{j\}}(x) - f_S(x)\big). \end{aligned} \tag{12}$$

The complete pipeline is outlined in Algorithm 1, ensuring accuracy, efficiency, and transparency in system. So, building on these gaps, we propose a generative discriminative IDS tailored for UAV networks. Traffic flows are first segmented and normalized (Eq. (2)–(3)), followed by a lightweight VAE that learns compact embeddings via reconstruction and KL divergence (Eq. (5), Eq. (6)). A supervised contrastive loss (Eq. (9)) further enforces class separation, leading to the joint GEM objective (Eq. (10)).

To address scarcity and imbalance, conditional GAN augmentation (Eq. (7)) synthesizes realistic attack traffic, which is classified using a CNN–GRU–Attention model under cross-entropy training. In our federated setup, the T-TIS dataset was partitioned across 10 virtual UAV clients, where each trained locally for 5 epochs before aggregation. Approximately 2 000 (1.8×) synthetic samples per attack class were generated.

For deployment, teacher networks are distilled and compressed (Eq. (8)) into a lightweight $\theta_{\text{compressed}}$ model. Inference relies on kernel-based anomaly scoring (Eq. (11)) with SHAP explanations (Eq. (12)) for interpretability.

IV. Experimental Setup

This section describes the experimental design, datasets, pre-processing pipeline, and evaluation metrics employed to validate the proposed GenAI-enabled UAV IDS framework.

This article has been accepted for publication in IEEE Transactions on Consumer Electronics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TCE.2026.3658881

Algorithm 1 Proposed GenFed-IDS: Generative Federated Learning Framework

Require: UAV Dataset (T-TIS) $X = \{x_1, \dots, x_N\}$, Labels Y, Clients K
Ensure: Intrusion label $\hat{y}$, Explanation report $\mathcal{E}$
// Phase 1: Preprocessing & Representation Learning
$X_{norm} \leftarrow$ Normalize(X) using Eq. (1)–(3)
Initialize VAE Encoder E_ϕ, Decoder D_θ
while not converged **do**
  Compute reconstruction loss $\mathcal{L}_{VAE}$ via Eq. (5)
  Compute regularization D_{KL} via Eq. (6)
  Compute contrastive loss $\mathcal{L}_{con}$ via Eq. (9)
  Update $\phi, \theta \leftarrow \nabla(\mathcal{L}_{VAE} + \lambda \mathcal{L}_{con})$
end while
// Phase 2: Generative Augmentation & Distillation
Train cGAN minimizing $\mathcal{L}_{GAN}$ via Eq. (7)
$Z_{syn} \leftarrow G(z|y)$ {Generate synthetic latent samples}
$\mathcal{D}_{aug} \leftarrow$ Latent(X_{norm}) $\cup Z_{syn}$
$\theta_{Teacher} \leftarrow$ TrainHybridModel($\mathcal{D}_{aug}$)
$\theta_{Student} \leftarrow$ Distill($\theta_{Teacher}$)
$\theta_{deploy} \leftarrow$ PruneAndQuantize($\theta_{Student}$) using Eq. (8)
// Phase 3: Federated Learning & Robustness
for round $r = 1$ to R **do**
  for client $k = 1$ to K **in parallel do**
    $\theta_k \leftarrow$ LocalUpdate($\theta_{deploy}, \mathcal{D}_k$)
  end for
  $\theta_{deploy} \leftarrow$ FedAvg($\{\theta_1, \dots, \theta_K\}$)
end for
$\theta_{final} \leftarrow$ AdversarialTrain(θ_{deploy}, FGSM/PGD)
// Phase 4: Inference (On-Device)
for new sample x_t **do**
  $z_t \leftarrow E_\phi(x_t)$ {Encode input}
  $score \leftarrow$ OC-SVM(z_t) using Eq. (11)
  if $score < 0$ **then**
    $\hat{y} \leftarrow$ Intrusion
    $\mathcal{E} \leftarrow$ SHAP(x_t, θ_{final}) using Eq. (12)
  else
    $\hat{y} \leftarrow$ Benign
  end if
  return $\hat{y}, \mathcal{E}$
end for=0

A. Simulation Environment

All experiments were conducted on a workstation equipped with an Intel Core i7-11800H (2.40 GHz, 8-core), 32 GB RAM, and an NVIDIA RTX 3070 GPU (8 GB VRAM). The implementation was carried out using Python 3.10 with PyTorch 2.0 and supporting libraries including Scikit-learn, Imbalanced-learn, LightGBM, and SHAP. Model quantization and pruning experiments were conducted using ONNX Runtime and TensorFlow Lite for embedded deployment. The complete software–hardware configuration is summarized in Table II.

TABLE II: System Specifications for Experimentation

System Aspect	Specification
CPU	Intel Core i7 (2.40 GHz, 8-core)
GPU	NVIDIA RTX 3070 (8 GB)
Operating System	Windows 11
RAM	32 GB
Language	Python 3.13
Libraries	PyTorch, Scikit-learn, Pandas, NumPy, SHAP

TABLE III: Details of the T-TIS Dataset

Category	Details
Total Samples	Cyber: ∼54,000; Physical: ∼45,000
Target Labels	0: Benign, 1: DoS, 2: Replay, 3: Evil Twin, 4: FDI
Benign	Cyber: 1–9426 Physical: 9427–13717
DoS Attack	Cyber: 13718–25389 Physical: 25390–26363
Replay Attack	Cyber: 26364–38370 Physical: 38371–39344
Evil Twin Attack	Cyber: 39345–45028 Physical: 45029–50502
False Data Injection (FDI)	Cyber: 50503–53976 Physical: 53977–54784
Total No. of Features	Cyber: 37 Physical: 16

B. Dataset

We utilized the *T-TIS* [25], which captures both cyber and physical telemetry of unmanned aerial vehicles under benign and adversarial conditions as mentioned in Table III. The dataset is collected through a UAV testbed comprising a drone, controller, and monitoring tools, where four major cyber-attacks were executed as dependent variable: de-authentication denial-of-service (DoS), replay attacks, false data injection (FDI), and evil twin attacks. It contains two complementary views as Independent variables: a cyber dataset with 37 features (e.g., packet statistics, protocol metadata, and timing information) and a physical dataset with 16 features (e.g., altitude, velocity, and positional data).

The dataset is well-organized and has distinct indices of benign and attack samples, as described in the original publication, and includes over 54,000 cyber and 45,000 physical records. In our experiments, we combined both cyber and physical modalities to create a multimodal feature space, so that the IDS could take advantage of cross-domain associations between network traffic and UAV flight behavior. To maintain the distribution of minority classes and increase resistance to overfitting, we used stratified k-fold cross-validation to divide the data into training, validation, and testing subsets.

C. Evaluation Metrics

The proposed UAV IDS performance is evaluated by standard classification measures, i.e. Accuracy (ACC), Precision (PR), Recall (RE), and F1-score (F1). These are based on the entries of the confusion matrix: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN).

The corresponding formulations are using Eq. 13 to 16:

$$ACC = \frac{TP + TN}{TP + TN + FP + FN}, \tag{13}$$

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TABLE IV: Model architecture and training configuration for UAV IDS.

Dataset	Input Features	Hybrid Modules	Dropout	Optimizer (lr)	Batch / Epochs†
UAV-IDS (T-ITS)	37	CNN (1D, 64) → MaxPool → GRU (128) → Autoencoder (128/64), MLP (64) (teacher); CNN (16) + GRU (32), Linear (student, pruned+quantized)	0.3 (linear/MLP), 0.0 (CNN/GRU)	Adam (1e-4)	64 / 30 (CV)/10 (student)

†Epochs: 30 per fold main/teacher, 10 for student, all with early stopping. Data standardized, 5-fold CV, post-training (std) with quantization and pruning.

TABLE V: Performance validation of the proposed model

Metric	Value
Accuracy	0.99488
Precision	0.99490
Recall	0.99488
F1-score	0.99488
AUROC	0.99999
Validation Loss	0.01415
FPR	0.00743
FNR	0.00521
MCC	0.99325
Cohen's Kappa	0.99428

$$PR = \frac{TP}{TP + FP}, \tag{14}$$

$$RE = \frac{TP}{TP + FN}, \tag{15}$$

$$F1 = \frac{2 \cdot PR \cdot RE}{PR + RE}, \tag{16}$$

Besides, Receiver Operating Characteristic (ROC) curves, Area Under the Curve (AUC), and Confusion Matrices (CM) are used to offer a holistic and comparative analysis.

TABLE VI: Confusion Matrix (Predicted vs Actual)

Predicted	Actual				
	Benign	DoS	Replay	Evil Twin	FDI
Benign	**2742.8**	0.00	0.20	0.00	0.20
DoS	0.00	**2511.4**	17.40	0.00	0.40
Replay	0.00	36.60	**2559.4**	0.00	0.20
Evil Twin	0.00	0.00	0.40	**2231.0**	0.20
FDI	0.00	0.00	0.40	0.00	**856.0**

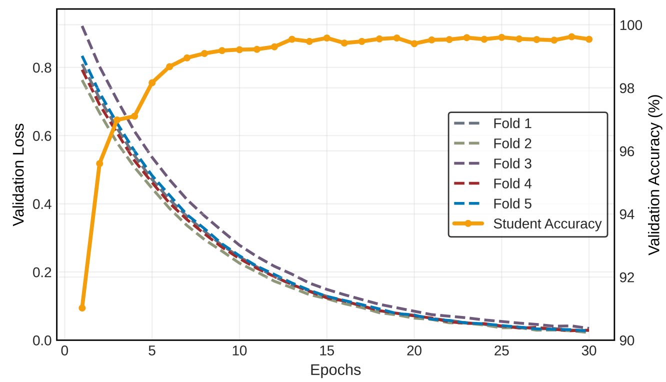

Fig. 3: Teacher model validation loss across 5 folds and the distilled student model's validation accuracy.

V. Results and Discussion

To test the performance of the proposed Generative AI-enabled IDS model, the UAV data was strictly tested with several evaluation metrics. All clients were trained locally in 5 epochs per round and global aggregation through FedAvg was done in 20 communication rounds. As summarized in Table V. The model had a very high accuracy of 99.49% per cent, precision, recall and F1-score values were all above 0.994, which means that the model has balanced and reliable classification. The model has a better discriminatory power between benign and malicious traffic as indicated by the AUROC of 0.999 Moreover, the low false positive rate (FPR = 0.00743) and false negative rate (FNR = 0.00521) prove its strength against both types of misclassification. The Matthews Correlation Coefficient (MCC = 0.99325) and Cohen Kappa (0.99428) support the consistency of predictions in different types of attacks, and the low validation loss indicates the consistency of model convergence.

The confusion matrix in Table VI gives more information on the classification results of the five traffic classes, which are Benign, DoS, Replay, Evil Twin, and False Data Injection. The diagonal dominance of the matrix indicates the near-perfect classification accuracy of all categories with only 0.2-0.4 cases per category being misclassified, mostly between Replay and DoS attacks. Notably, the model was able to identify all the cases of Evil Twin and FDI attacks with insignificant error rates, which proves the effectiveness of the model in identifying advanced threats. All in all, these findings confirm the capability of the model to provide very accurate, reliable, and secure intrusion detection in real-life UAV communication conditions.

In order to further evaluate the learning behavior, Fig. 3 shows the validation loss curves of the 5-fold cross-validation of the teacher model.

The loss consistently decreases with epochs, confirming smooth convergence and stable generalization across all folds. In parallel, the same figure presents the validation accuracy of the distilled student model. The model rapidly improves within the first few epochs and stabilizes above 99% accuracy, highlighting the effectiveness of knowledge distillation in achieving lightweight yet highly accurate intrusion detection.

To interpret the decision-making of the proposed IDS, SHAP analysis was used to identify the most important features across all attack classes. Fig. 4 shows the top 10 features contributing to the classification of benign and malicious traffic. Features such as `ip.id_x`, `data.len_x`, and `frame.len_x` had the highest impact on the model output, showing their key role in distinguishing between normal and abnormal UAV communications.

The color-coded bars represent contributions for each attack category as defined in the confusion matrix (Class 0: Benign, Class 1: DoS, Class 2: Replay, Class 3: Evil Twin, Class 4: FDI). The results indicate that while some features (e.g., `ip.id_x`) are dominant across multiple classes, others (e.g., `tcp.window_size_x` and `wlan.duration_x`) provide

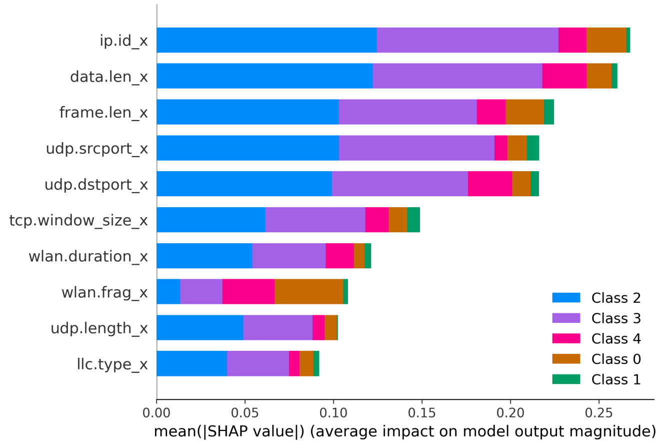

Fig. 4: SHAP summary plot of the top 10 most influential features for benign and attack classes.

TABLE VII: Model performance and lightweight capacity comparison between Teacher and Student models.

Attribute	Teacher Model	Student Model	Unit
Model Size	6.5	3.2	MB
Parameters	1.58	0.86	Million
Training Time	0.85	0.43	s/epoch
Total Training Time	5.4	3.6	min
Inference Latency	0.83	0.41	ms/sample
GPU Memory Usage	64	37	MB

class-specific insights. This shows that the model not only achieves high accuracy but also learns meaningful feature representations relevant to UAV intrusion detection.

Further, Table VII compares the performance and lightweight capacity of the teacher and student models. The results show that the student model is almost half the size of the teacher model (3.2 MB vs. 6.5 MB) with fewer parameters (0.86M vs. 1.58M). Training and inference are also faster for the student model, with training time per epoch reduced from 0.85s to 0.43s and inference latency reduced from 0.83ms to 0.41ms per sample. In addition, the student model consumes less GPU memory (37 MB compared to 64 MB). These results confirm that the knowledge-distilled student model achieves a much lighter footprint while maintaining competitive performance.

Finally, to highlight the efficiency gains and novelty, a comparative analysis between the proposed model and baseline IDS models was conducted in terms of the standard evaluation metrics along with model size, computational cost, and memory footprint. As shown in Fig. 5, the proposed CNN-GRU-AE framework significantly outperforms recent state-of-the-art IDS models, including GAN-MCNN-DFS, ANN, CNN, XAI-DNN, and SEMI-GRU.

In particular, the proposed model achieved the highest accuracy (99.49%), precision (99.49%), recall (99.48%), and F1-score (99.48%), surpassing the closest baseline (GAN-MCNN-DFS) by margins of 4–7%. The recall improvement is especially important for UAV network security, as it minimizes the chances of undetected attacks. While achieving superior detection capability, the distilled student model also demonstrated lightweight characteristics with reduced model size, faster training and inference times, and lower GPU memory usage.

These findings prove that the suggested framework does not only promote the accuracy of detection but also fulfills the essential need of resource efficiency, which is very appropriate in UAV communication networks.

VI. Conclusion

This paper has suggested a Generative AI-based lightweight intrusion detection system that is specific to UAV communication networks. The model combines hybrid CNN-GRU autoencoders with knowledge distillation to obtain high detection accuracy and low computational overhead. The strength of the framework was proven by extensive experiments with an accuracy, precision, recall, and F1-score of 99.49% and almost perfect values of the AUROC. The confusion matrix proved that there was little misclassification between benign and attack classes, which proved the effectiveness of the proposed approach.

Cross-validation revealed that the convergence was stable, and the SHAP-based explainability demonstrated the most significant features that influenced the classification decisions. The distilled student model was very resource-efficient, decreasing model size, training and inference time, and GPU memory consumption. In addition, the comparative analysis with the current baseline IDS models showed that the proposed framework was superior in terms of performance and efficiency.

The proposed IDS framework offers an effective and practical solution for securing UAV networks against evolving threats. In the future, we plan to extend this work by incorporating real-time deployment on UAV platforms and exploring reinforcement learning-based adaptive defense strategies.

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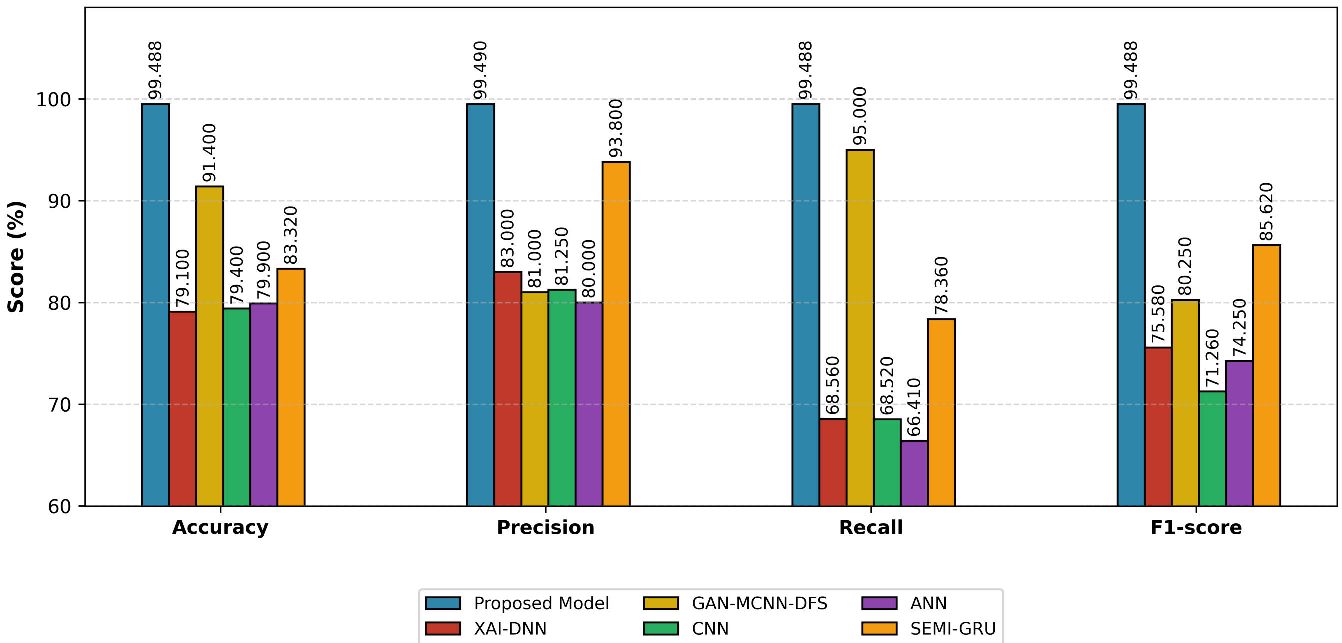

Fig. 5: Comparison of the proposed framework with baseline IDS models from recent studies across performance metrics

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